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Key Points:

- Temperature had an inverted U-shape with respiratory syncytial virus infections, while relative humidity was positively correlated
- The risk effects of extreme climatic factors were observed on the concurrent day and persisted over extended lag periods
- The attributable burden of respiratory syncytial virus infections associated with cold was higher than that related to humid conditions

Supporting Information:

Supporting Information may be found in the online version of this article.

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Association of Ambient Temperature and Relative Humidity With Respiratory Syncytial Virus Infections Among Hospitalized Children in Suzhou, Eastern China: A Time-Series Analysis

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Abstract Respiratory syncytial virus (RSV) is the leading cause of clinical pneumonia in children. We aimed to investigate the associations between ambient temperature, relative humidity, and pediatric RSV infections, and to assess the disease burden attributable to cold or humid conditions. Daily data on RSV hospitalizations among children aged ≤ 5 years, mean temperature, and relative humidity in Suzhou, China, from January 2016 to December 2019 were collected. A distributed lag nonlinear model with quasi-Poisson regression was employed to assess the exposure-lag-response associations. Attributable risks were calculated to quantify the disease burden due to climatic factors. We found an inverted U-shaped relationship between temperature and RSV infections, with the cumulative risk of RSV peaking at 7.5°C (RR = 4.30, 95% CI: 3.08–6.02). The exposure-response curves for relative humidity exhibited a generally positive trend, peaking at 100.0% (RR = 3.14, 95% CI: 1.84–5.34). Using median values as references, the highest risk effects of extremely low (RR = 1.14, 95% CI: 1.04–1.25) and low (RR = 1.22, 95% CI: 1.12–1.32) temperatures, as well as high (RR = 1.09, 95% CI: 1.04–1.13) and extremely high (RR = 1.16, 95% CI: 1.07–1.27) relative humidity, occurred on the day of exposure and persisted for extended periods. The attributable fraction of RSV infections associated with cold or humid conditions was 55.23% (95% CI: 50.01%–64.03%) and 12.02% (95% CI: 9.36%–20.24%), respectively. The risk effect of high relative humidity was stronger in children aged 1–5 years. Our findings suggest nonlinear, lagged associations between climatic factors and pediatric RSV infections, which may inform future healthcare planning and RSV immunization strategies.

Plain Language Summary Respiratory syncytial virus (RSV) is the leading pathogen responsible for clinical pneumonia in infants and young children. RSV epidemics typically occur during the winter in temperate and subtropical climates, whereas in tropical climates, peak activity is observed during the rainy season. Climatic variables play a key role in shaping RSV seasonality, and understanding these drivers is essential for enhancing epidemic forecasting, informing healthcare planning, and developing effective prevention strategies. In this study, we used a distributed lag non-linear model to investigate the exposure-lag-response associations between ambient temperature and relative humidity and pediatric RSV infections in Suzhou, Eastern China. From 2016 to 2019, a total of 5,008 hospitalized cases of RSV infections among children aged ≤ 5 years were included in this analysis. We observed an inverted U-shaped relationship between ambient temperature and RSV infections, while relative humidity exhibited a generally positive correlation. The risk effects associated with extreme conditions of low temperature and high relative humidity were observed on the day of exposure and extended over prolonged lag periods. Besides, cold or humid conditions could account for a substantial proportion of pediatric RSV infections. These findings will enhance early warning systems for RSV epidemics and inform healthcare planning.

1. Introduction

Respiratory syncytial virus (RSV) is the leading pathogen responsible for acute lower respiratory infections in infants and young children, with RSV infections ranking as the second most frequent cause of infant mortality globally (Lozano et al., 2012; Terstappen et al., 2024). Serological surveys have indicated that nearly every child is likely to be infected at least once with RSV before reaching 5 years of age (T. T. Lam et al., 2019). RSV imposes a significant burden on healthcare systems. Globally, it is estimated to lead to approximately 3.6 million hospital